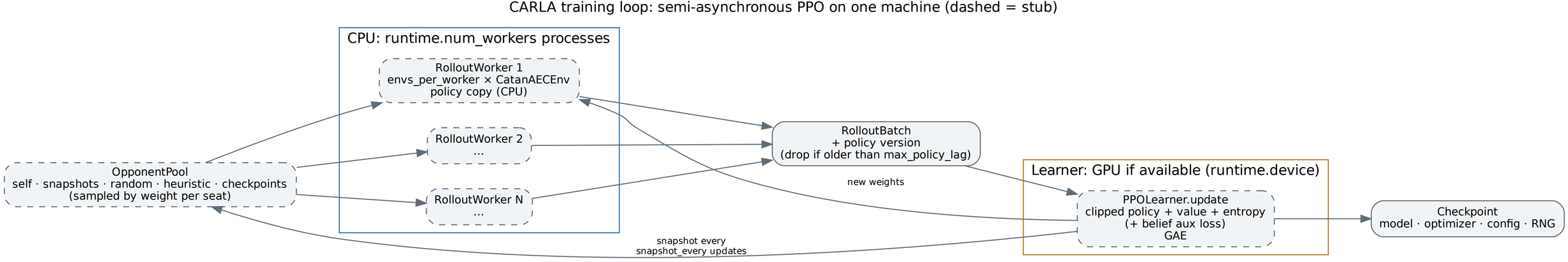
CARLA training loop: semi-asynchronous PPO on one machine (dashed = stub)
CPU: runtime.num_workers processes
RolloutWorker 1
envs_per_worker × CatanAECEnv
policy copy (CPU)
RolloutWorker 2
...
RolloutWorker N
...
OpponentPool
self · snapshots · random · heuristic · checkpoints
(sampled by weight per seat)
RolloutBatch
+ policy version
(drop if older than max_policy_lag)
new weights
Learner: GPU if available (runtime.device)
PPOLearner.update
clipped policy + value + entropy
(+ belief aux loss)
GAE
Checkpoint
model · optimizer · config · RNG
snapshot every
snapshot_every updates